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Roll No

FT-503 (GS)

B.E. V Semester Examination, June 2020

Grading System (GS)

Machine Drawing and Design

Time : Three Hours

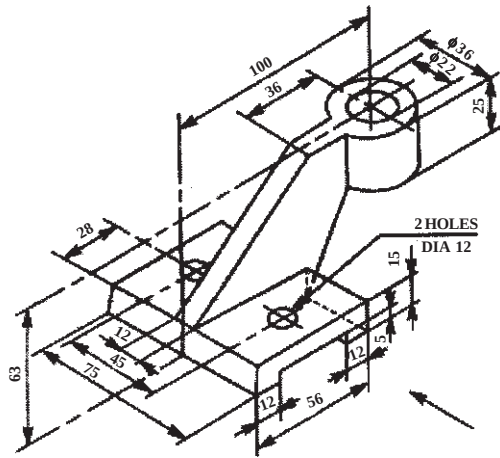
Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe:
 - i) Wire frame modelling
 - ii) Surface modelling
 - iii) Solid modellingb) Name at least seven edit command used in Auto CAD and also discuss their functions.
2. a) What do you mean by sectional views? What are its different types?
b) Discuss two methods dimensioning, normally use in dimensioning.
3. Discuss the general procedure adopted in solving a design problem and show them on a flow chart.
4. A triple riveted lap joint is to be made between 6.5 mm plates. The allowable stresses $f_t=35\text{N/mm}^2$, $f_s=29\text{N/mm}^2$, $f_c=52.5\text{ N/mm}^2$. Calculate the rivet diameter, rivet pitch and back pitch, zig-zag riveting is to be used.
5. Design a Knuckle Joint for a tie rod of a circular section to sustain maximum pull of 70kN. The ultimate tensile strength of rod material is 420N/mm^2 . The ultimate tensile strength and shearing strength of pin material are respectively 720N/mm^2 and 395N/mm^2 . Use factor of safety 6.
6. Explain basic design procedure of machine elements. What is the different design considerations used in designing machine elements?

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7. Draw front view, top view and side view of the following figure.



8. Define:(Any two)

- i) Roughness
- ii) Surface Roughness No.
- iii) Tolerance
- iv) Allowance
